11/09/24

Algorithm (PB)

Ford Fulkerson Algorithm Analysis

① $O(F^{*}m)$ max flow value
↳ constant

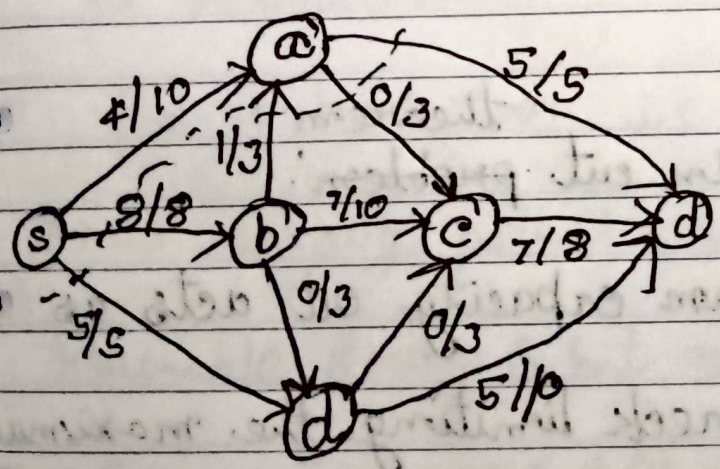
② Gives the optimum

Obs^n: ① f, f'

$|f| + |f'|$ valid flow

② netflow from X and Y

$$f(X,Y) = \sum_{x \in X} \sum_{y \in Y} f(x,y) - \sum_{y \in Y} \sum_{x \in X} f(y,x)$$



$f(x,y) = 17$

$f = 17$

$X = \{s, a\}$

$Y = \{b, c, d, t\}$

Lemma: Let X, Y be any s-t cut, Given any flow f, the value of f is equal to net flow.

Capacity of a cut (X, Y): $C(X,Y) = \sum_{u \in X} \sum_{v \in Y} c(u,v)$

Sum of capacities of Edges leaving from X and Y

Lemma: Given any s-t cut (X, Y) and any flow, the value of the flow value is $|f| \leq C(X,Y)$

Theorem:

Max flow min cut ~~problem~~ theorem:

[minimum capacity cut acts as a bottleneck limiting the maximum amount of flow]

i) f is a maximum flow in G

ii) The residual network G_f contains no

aug path

iii) $|f| = c(X,Y)$ for some cut (X,Y) of G

i) -> ii) by contradiction

statement i but there is still a path

=> we can push more flow

=> f is not max

=> contradiction

the contradiction is not true

SUNDAY 09

ii) -> iii) no aug path => some vertices are

reachable from s but not t and come

other vertices => cut (X,Y) => $c(X,Y) = |f|$

(previous lemma)

iii) ⇒ i) contradiction

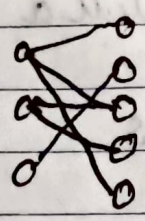
Suppose there is a flow f' & $|f'| > |f|$

$\Rightarrow |f'| > c(X,Y)$ ~~[from lemma $|f| \leq c(X,Y)$]~~

this contradicts the lemma $|f| \leq c(X,Y)$

(proved)

Bipartite graph $G = (X \cup Y, E)$

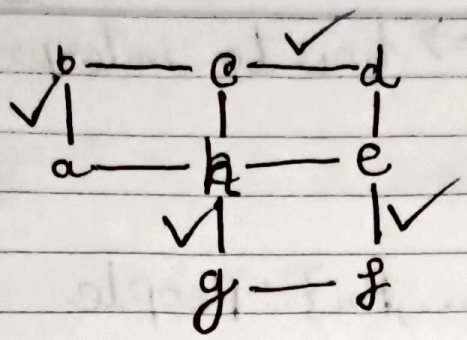


$(u,v) \in E,\ u \in X,\ v \in Y$

Maximum bipartite matching

Matching: Given $G: (V,E)$

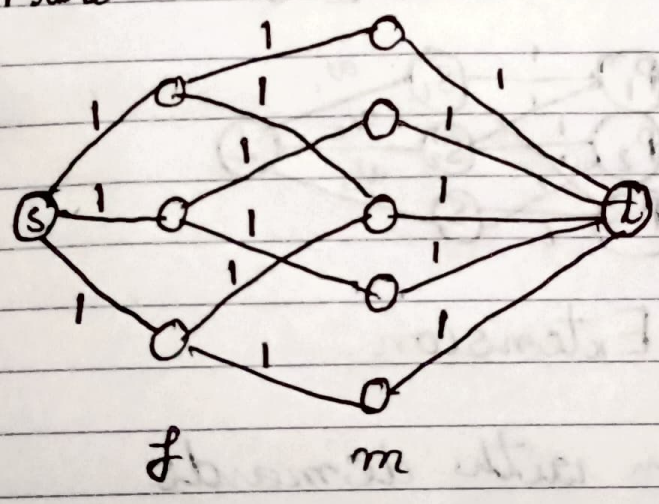
given a set of edges M such that for each $v \in V$ there is at most one edge of M incident to v



matching set = $\{(a,b),(c,d),(e,f),(g,h)\}$

$|M| = 4$

Reduction of maximum bipartite graph to Flow network:



creation of network N

① set the vertices

② set the edges

③ capacities of edges

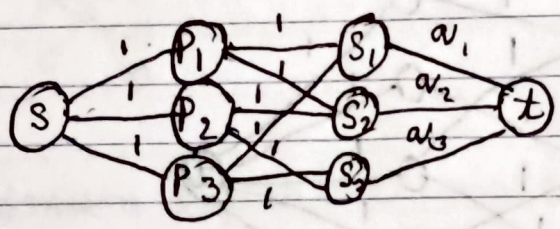
④ Max flow (FF) ⇒ how to interpret the value of flow

people $P = \{P_1, P_2, \ldots, P_n\}$: People

$S = \{S_1, S_2, \ldots, S_m\}$: Spot

$Q = \{q_1, q_2, \ldots, q_m\}$. quantity of people at the spot

P_1 can visit → $S_{P_1} \subseteq S$



Circulation: Extension

Circulation with demands:

Given Directed Graph $G = (V, E)$

1) $(u,v) \in E$ & $c(u,v)$: capacity of edge (u,v)

2) With each vertex we have supply/demand

value d_v

supply $\Rightarrow d_v < 0$; supply nodes: S

demand $\Rightarrow d_v > 0$; demand nodes: T

else other vertex $\Rightarrow d_v = 0$

Q: does there exist a valid circulation?

get a Reduction.

capacity constraint: $(u,v) \in E$, $f(u,v) \leq c(u,v)$

Supply demand: